

The
non-experimental
evaluation problem

Typical sources of
endogeneity

Randomized Control
Trial (RTC)

Main idea:
make treated
and untreated
alike

Confounding and
Controls

Natural Experiments

Difference-in-
Differences
(DiD)

Difference-in-
Differences (DiD)

Example: Mariel
Boatlift (Nat. exp.
& DiD)

Instrumental
Variables (IV)

IVs & 2-stage least
squares

Example: Settler

ECONOMETRICS I

Lecture 10

Treatment evaluation: an overview

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The non-experimental evaluation problem

Problem: evaluate the impact β_1 of “treatment” x_1 on the “outcome” y through the regression

$$y_i = x_{1i}\beta_1 + x_{2i}\beta_2 + \dots + u_i$$

where x_1 may be a dummy (if the treatment is binary) or a continuous intensity variable.

Critical assumption:

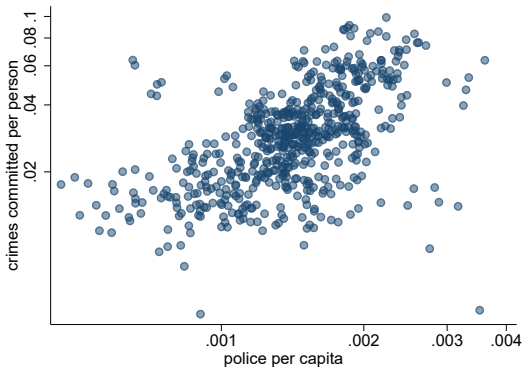
$$\mathbb{E}(u|X) = 0 \implies \boxed{\text{cov}(u_i, x_i) = 0}$$

Exogeneity

The non-experimental evaluation problem

Slides 10

Does the police reduce crime?

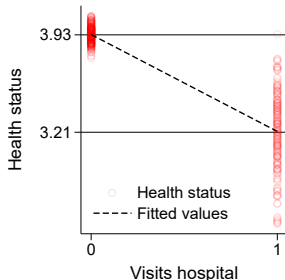


The non-experimental evaluation problem

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Are hospitals good for health?

Group	Sample Size	Mean Health Status	Std. Error
Hospital	7,774	3.21	0.014
No hospital	90,049	3.93	0.003



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Problem: reverse causality (self-selection)

- hospital $\rightarrow$ health $\rightarrow$ hospital $\rightarrow$ health $\rightarrow \dots$
- police $\rightarrow$ crime $\rightarrow$ police $\rightarrow$ crime $\rightarrow \dots$
- price $\rightarrow$ sales $\rightarrow$ price $\rightarrow$ sales $\rightarrow \dots$

$$Y = X\beta + u$$

$$X = Y\gamma + v$$

$u \rightarrow X$ means $\boxed{\text{cov}(u_i, x_i) \neq 0}$
Endogeneity

Typical sources of endogeneity

Endogeneity 1: Simultaneous causality

Research question: Effect of police on crime?

$$\text{Crime} = \beta_1 + \beta_2 \text{Police} + \epsilon \quad (\beta_2 < 0) \quad (1)$$

But reverse causality also exists:

$$\text{Police} = \alpha_1 + \alpha_2 \text{Crime} + \nu \quad (\alpha_2 > 0) \quad (2)$$

Simultaneity: Police causes Crime AND Crime causes Police.

Solving the system gives:

$$\text{Crime} = \frac{\beta_1 - \alpha_1}{1 + \alpha_2} + \frac{1 + \beta_2}{1 + \alpha_2} \text{Police} + \frac{\epsilon - \nu}{1 + \alpha_2} \quad (3)$$

OLS estimates the slope of (3) instead of that in (1).

Typical sources of endogeneity

Endogeneity 2: Omitted variable bias

Research question: Effect of education on income?

$$\text{Income} = \hat{\beta}_1 + \hat{\beta}_2 \text{Education} + \hat{u} \quad (4)$$

But true data generating process (DGP):

$$\text{Income} = \beta_1 + \beta_2 \text{Education} + \beta_3 \text{Talent} + \nu \quad (5)$$

Problem: If Education and Talent are correlated (talented people study more), then:

- $u = \beta_3 \text{Talent} + \nu$ correlates with Education
- $\mathbb{E}[X'u] \neq 0$
- OLS estimate $\hat{\beta}_2$ is biased

Typical sources of endogeneity

Endogeneity 3: Measurement errors

Research question: Effect of historical inequality (x) on current development (y)?

True model:

$$y = \beta_1 + \beta_2 x + \epsilon \quad (6)$$

But we observe mismeasured variable:

$$\tilde{x} = x + \delta \quad (\delta = \text{measurement error}) \quad (7)$$

We incorrectly estimate:

$$y = \hat{\beta}_1 + \hat{\beta}_2 \tilde{x} + \hat{u} \quad (8)$$

Problem: Error term becomes $u = \epsilon - \beta_2 \delta$, which correlates with $\tilde{x}$.

Hence: $\mathbb{E}[u|X] \neq 0$.

Typical sources of endogeneity

How can we avoid $\text{cov}(u_i, x_i) \neq 0$?

Popular solutions:

- 1 Gold standard: **Randomized Controlled Trial (RCT)**
(Other methods will try to replicate this ideal)
- 2 **Natural experiments**/accidents
 - with **difference in difference (DiD)**
 - with **instrumental variable (IV)** independent of u_i
- 3 Almost treated vs. almost untreated: **Regression discontinuity designs (RDD)**
- 4 Overall idea: neutralize differences except treatment
 - Control for confounders (e.g., fixed effects, DiD)
 - Matching (similar treated/untreated)

Randomized Control Trial (RTC)

Idea: Random assignment ensures treatment is independent of potential outcomes.

Setup:

- Treatment randomly assigned (like in medical trials).
- No selection bias: $\mathbb{E}[u \mid x] = 0$.
- Estimate causal effect via difference in means.

Examples

- Randomly give patient medicine or placebo.
- Assigning police to neighborhoods at random.
- Price your product after rolling a dice.

Limitations: Costly, sometimes unethical or unfeasible.

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MAIN IDEA: MAKE TREATED AND UNTREATED ALIKE

Confounding and Controls

Reminder

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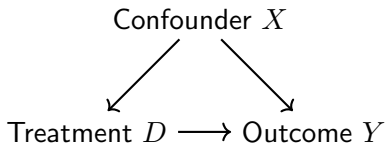
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- In observational data, treatment may be related to other variables that also affect outcome.
- We call these **confounders**.
- Controlling for them in regression helps isolate the treatment effect.



- **Important even under randomization!**

Confounding and Controls

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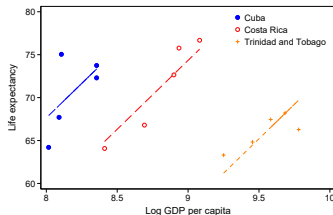
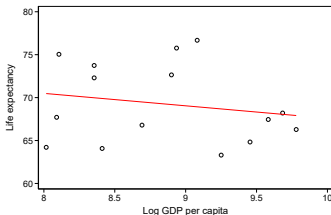
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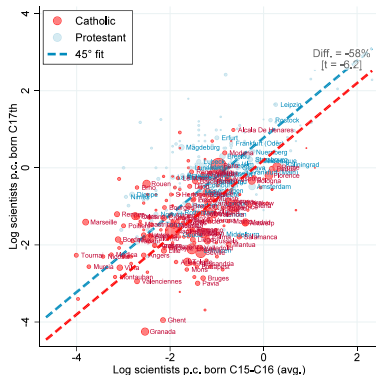
Recall: with and without fixed effects



FE by country capture omitted variable and reveal the positive slope.

Confounding and Controls

Example: Did Catholic reactionary isolationism affect science?



Catholic dummy associated with fewer famous scientists per capita. But is it causal?

Confounding and Controls

Did Catholic reactionary censorship affect science?

	Dependent variable: Change in scientists p.c. from 16th to 17th century								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	All cities	Religious border (100km)	All cities	Nearby Catholics (100km)	Iberia	Italy	All cities	Dutch border (100km)	All cities
<u>Counter-Ref. intensity</u>									
Catholic ruler 1600	-0.59***	-0.73***	-0.18				-0.24*		0.04
Strictly Catholic regime			-0.48***				-0.33***		-0.32**
Inquisitorial tribunal			-0.40***	-0.36***	-0.51*	-0.36***	-0.46***		-0.37***
Spanish Empire 1600						-0.61***	-0.35***	-0.55***	-0.36***
Catholic university 1700									-0.53***
Catholic pop. share [0,1]									-0.44***
<u>Controls</u>									
University in 1700	0.18**	0.35*	0.24***	-0.05	0.12	-0.07	0.25***	0.43***	0.58***
Dist. to Atlantic	✓	✓	✓	✓	✓	✓	✓	✓	✓
Dist. to Mediterranean	✓	✓	✓	✓	✓	✓	✓	✓	✓
Initial researchers p.c.	✓	✓	✓	✓	✓	✓	✓	✓	✓
Printing volume (by 1500)	✓	✓	✓	✓	✓	✓	✓	✓	✓
Pop. growth 1600–1700	✓	✓	✓	✓	✓	✓	✓	✓	✓
Pop. growth 1500–1700	✓	✓	✓	✓	✓	✓	✓	✓	✓
Ethno-linguistic FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Fluv. catchment region FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	649	172	649	171	43	105	649	75	641
R-squared	0.46	0.50	0.48	0.51	0.55	0.53	0.49	0.40	0.51

Notes: Spatial-autocorrelation-adjusted standard errors within 200 km based on Colella et al. (2019). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Natural Experiments

Idea: Exploit events or rules that mimic random assignment.

Setup:

- Assignment driven by external factors (e.g., accidental policy interruption).
- Credibility hinges on “as-good-as-random” variation.
- Often analyzed via Difference-in-Differences or IV.

Example: Draft lottery number as exogenous variation in military service.

Limitation: Validity depends on plausibility of the exogeneity assumption.

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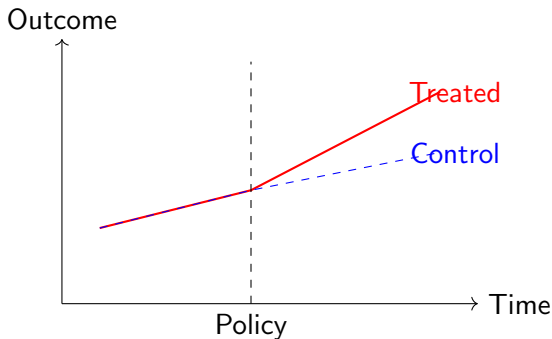
Example: Settler

DIFFERENCE-IN-DIFFERENCES (DiD)

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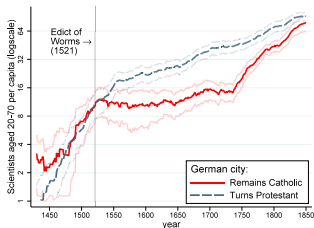
Idea

- Compares pre-post differences between treated and control groups.
- Key assumption: Parallel trends.

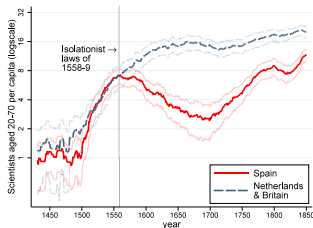


Difference-in-Differences (DiD)

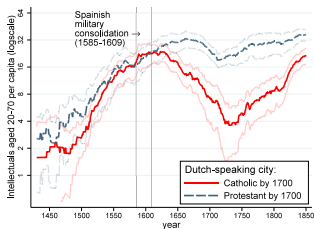
(a) German-speaking cities



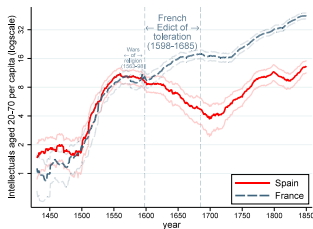
(b) Spain vs. Britain and the Netherlands



(c) Dutch-speaking cities



(d) France's special case

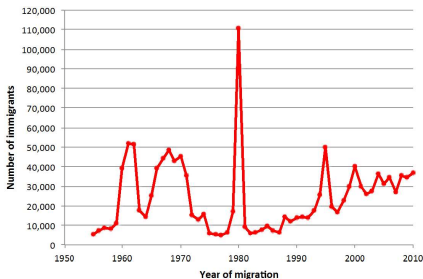


Example: Mariel Boatlift (Nat. exp. & DiD)

Research Question: How do wages react to immigration?

Natural experiment: Mariel Boatlift of 1980

- April-October 1980: 125,000 Cuban immigrants arrived in Miami
- Miami labor force increased by 7% almost overnight



Example: Mariel Boatlift (Nat. exp. & DiD)

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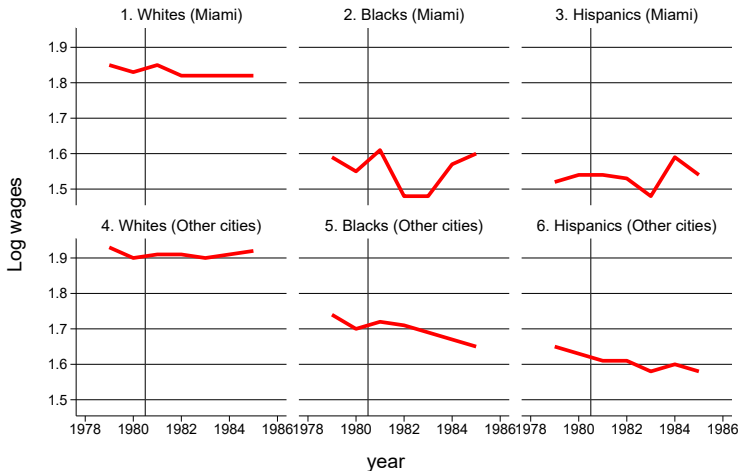
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Example: Mariel Boatlift (Nat. exp. & DiD)

David Card's (1990) methodology:

- Difference-in-differences design
- Compared Miami to similar "control cities" (Atlanta, Houston, etc.)
- Examined wages of low-skilled workers (high school dropouts)
- Pre-period: 1979, Post-period: 1985

Findings:

- **No significant effect** on wages of low-skilled workers
- Unemployment rates unaffected
- Suggests local labor markets absorb immigration shocks

Impact: Became the canonical finding for decades

Example: Mariel Boatlift (Nat. exp. & DiD)

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Card (1990): No Effect

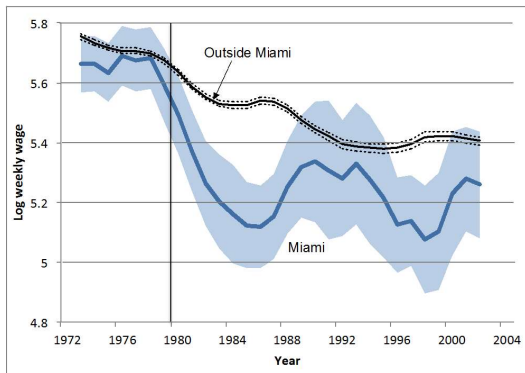
- Difference-in-differences: Miami vs. other cities
- Found no significant wage changes for low-skilled workers
- Became canonical result for decades

Borjas (2017): Large Negative Effect

- Critique: Card used wrong comparison groups
- New approach: compare similar workers within Miami
- Found 10-30% wage drops for high school dropouts

Example: Mariel Boatlift (Nat. exp. & DiD)

**Figure 2. Log wage of high school dropouts, 1972-2003
(95 percent confidence band)**



Example: Mariel Boatlift (Nat. exp. & DiD)

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Clemens & Hunt (2019): zero effect confirmed

- Critique: Borjas had small samples, high variance, driven by population subsamples unrelated to Cuban migration.
- Current consensus: market absorbed shock successfully

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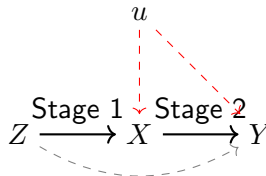
IVs & 2-stage least
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Example: Settler

INSTRUMENTAL VARIABLES (IV)

IVs & 2-stage least squares

IV Intuition: The Two-Stage Idea



Stage 1: Regress endogenous X on instrument Z :

$$X = Z\pi + v$$

Get predicted values: $\hat{X} = Z\hat{\pi}$

Stage 2: Regress y on $\hat{X}$:

$$y = \hat{X}\beta + \epsilon$$

IVs & 2-stage least squares

IV Assumptions

1 **Relevance:** $\text{cov}[Z, X] \neq 0$ (Z correlated with X)

2 **Exogeneity:** $\text{cov}[Z, u] = 0$ (Z uncorrelated with X)

1 + 2 = **exclusion restriction:** Z affects y only through X

Properties:

- **Consistent:** $\hat{\beta}_{2SLS} \xrightarrow{p} \beta$ (more next class ...)
- **Asymptotically normal** under regularity conditions
- **Less efficient** than OLS (when OLS is valid)
- Standard errors must account for two-stage estimation

Weak instrument problem: If Z weakly correlated with X , IV can be badly biased

IVs & 2-stage least squares

Example: Returns to Education

Model: $\log(wage) = \beta_0 + \beta_1 education + u$

Problem: Ability bias (u contains unobserved ability)

Potential Instruments:

- Distance to college (Card 1995)
- Randomized Vietnam War draft (exclusion restriction satisfied?)

Intuition: These affect education but not wages directly (sure?)

Caution: Each instrument requires careful exclusion restriction argument

Example: Settler mortality, institutions and growth (Nat. exp. & IV)

Research Question: What determines long-run economic development?

- Why are some countries rich and others poor?
- **Colonial origins** hypothesis: Historical institutions shape modern development
- Key mechanism: Settler mortality → settlement patterns → institutions → development
- Acemoglu, Johnson & Robinson (2001): Founding paper

Example: Settler mortality, institutions and growth (Nat. exp. & IV)

Acemoglu-Johnson-Robinson (2001) Methodology

- Instrumental variables approach:
- **Stage 1:** Settler mortality → Early institutions → Current institutions
- **Stage 2:** Early & current institutions → Modern GDP
- Data: Historical mortality rates of European settlers
- Sample: Former European colonies

Findings:

- High mortality → Few settlers → Extractive institutions
- Low mortality → Many settlers → Inclusive institutions
- Persistent effects on modern institutions and income

Example: Settler mortality, institutions and growth (Nat. exp. & IV)

AJR empirical approach

First stage:

$$\text{Institutions}_i = \alpha + \beta \log(\text{Settler Mortality})_i + \epsilon_i$$

Second stage:

$$\log(\text{GDP pc})_i = \gamma + \delta \widehat{\text{Institutions}}_i + u_i$$

Their Result: Strong, significant relationship

- 1 SD increase in mortality $\rightarrow$ 1.7 SD worse institutions
- Institutions explain bulk of income differences
- **Do you believe this?** (think about the exclusion restriction: what if settler brought not just institutions but also trading networks, technical knowledge, etc.)

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REGRESSION DISCONTINUITY DESIGN (RDD)

RDD: Returns to Elite University Education

RDD: Intro

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Hoekstra (2009): "The Effect of Attending the Flagship State University on Earnings"

Research Design: Regression Discontinuity using admission cutoffs at Florida State University

- Compare students just above vs. below SAT cutoff
- Local randomization around admission threshold

Findings:

- Large earnings premium of marginal admitted students
- Flagship university attendance has significant labor market returns

RDD: Returns to Elite University Education

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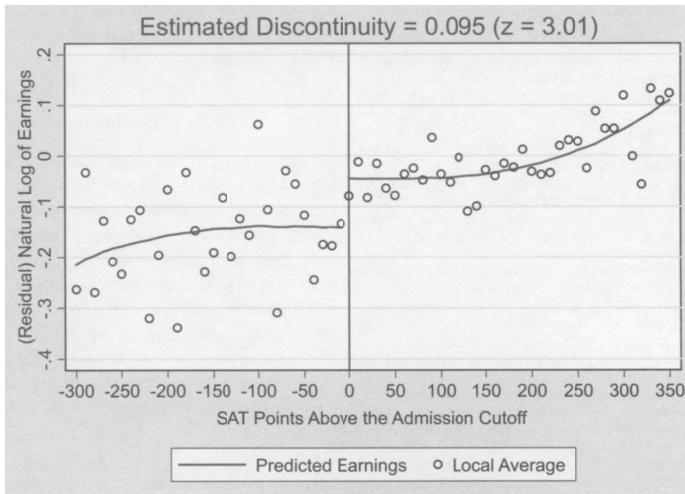
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RDD: Returns to Elite University Education



Journal of Econometrics
Volume 142, Issue 2, February 2008, Pages 675-697



Randomized experiments from non-random selection in U.S. House elections ☆

David S. Lee

From the abstract:

“... the inherent uncertainty in the final vote count is plausible, which would imply that the party that wins is **essentially randomized** among elections decided by a narrow margin. The evidence is consistent with this prediction, which is then used to generate “near-experimental” causal estimates of the electoral advantage to incumbency.”

RDD: Returns to Elite University Education

Randomized

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- Running variable: Democratic margin of victory in election t :
$$\text{margin}_t = \text{votes}_{\text{Dem}} - \text{votes}_{\text{Rep}}$$
- Treatment: Democrat barely wins ($\text{margin}_t > 0$)
- Outcome: Democrat wins again in $t + 1$
- Estimand: Local Average Treatment Effect (LATE) at the cutoff.

RDD: Returns to Elite University Education

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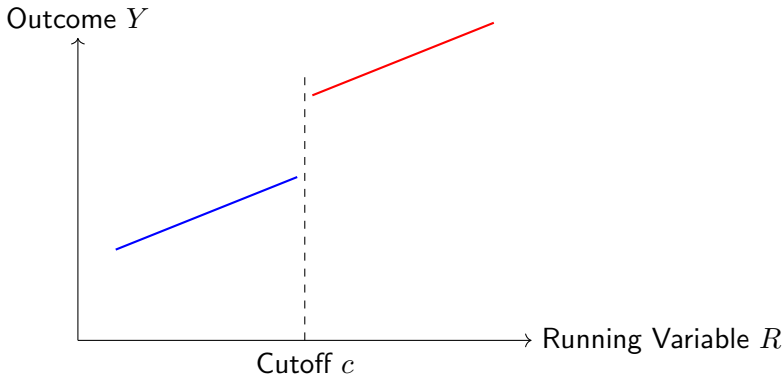
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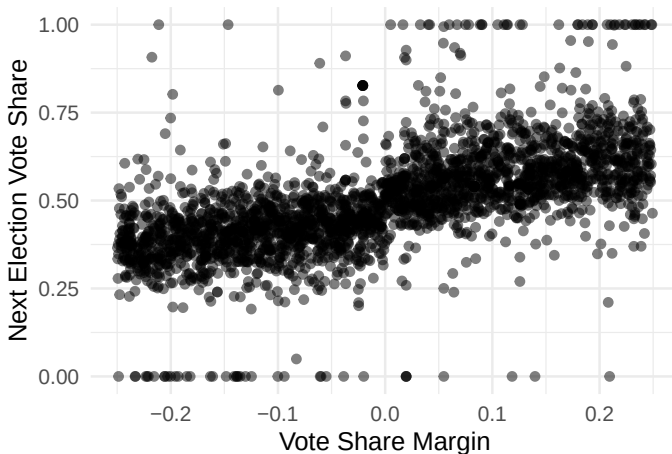
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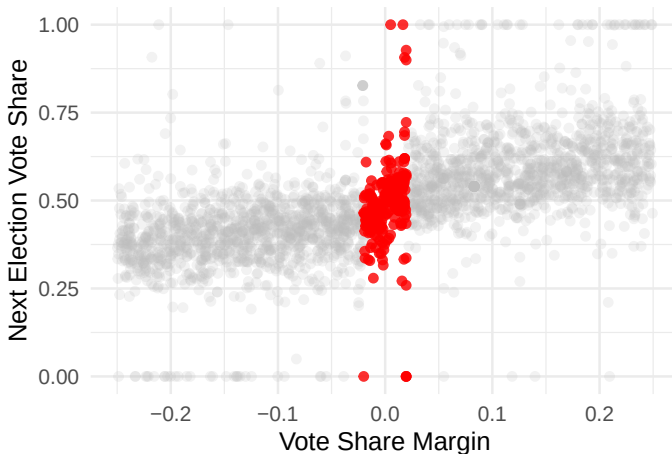
RDD: Returns to Elite University Education

All Data: Vote Share vs. Next Vote Share



RDD: Returns to Elite University Education

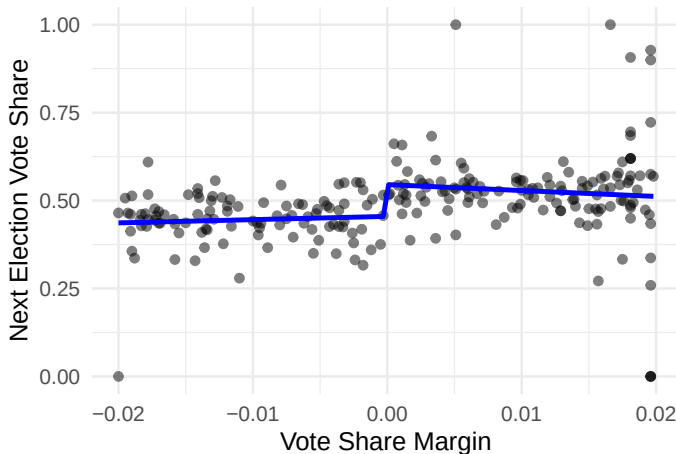
Highlighting Close Elections ($|\text{share}| = 0.02$)



RDD: Returns to Elite University Education

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RDD Estimate Around Cutoff



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CONCLUSIONS

Recap

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